

3、雷达避障趣味玩法

该章节需要搭配小车底盘以及其它传感器才能运行，这里只是说明实现的方式，实际上运行不了，需要搭配本公司的RDK-X3旭日派小车实现该功能。要移植到自己的主板上运行，还需安装其它依赖，因此，这里只是做代码演示，望须知。

3.1、功能说明

程序启动后，小车会往前走，当检测范围内出现障碍物的时候，会调整姿态，避开障碍物，随后继续往前走。

3.2、代码参考路径

该功能源码的位置位于配套虚拟机中，

```
~/oradar_ws/src/yahboomcar_laser/yahboomcar_laser/laser_avoidance.py
```

3.3、程序启动

终端输入，

```
ros2 launch yahboomcar_laser laser_avoidance_launch.py
```

3.4、核心代码解析

laser_avoidance_launch.py

```
from launch import LaunchDescription
from launch_ros.actions import Node

import os
from ament_index_python.packages import get_package_share_directory
from launch.actions import IncludeLaunchDescription
from launch.launch_description_sources import PythonLaunchDescriptionSource

def generate_launch_description():
    ms200_scan_node = IncludeLaunchDescription(
        PythonLaunchDescriptionSource([os.path.join(
            get_package_share_directory('oradar_lidar'), 'launch'),
            '/ms200_scan.launch.py'])
    )

    laser_avoidance_node = Node(
        package='yahboomcar_laser',
        executable='laser_avoidance',
        output = 'screen'
    )

    bringup_node = IncludeLaunchDescription(
        PythonLaunchDescriptionSource([os.path.join(
            get_package_share_directory('yahboomcar_bringup'), 'launch'),
```

```

        '/yahboomcar_bringup_launch.py''])
    )

    launch_description = LaunchDescription([
        ms200_scan_node,
        laser_avoidance_node,
        bringup_node
    ])
    return launch_description

```

主要是启动雷达ms200_scan_node、底盘bringup_node和避障功能节点laser_avoidance_node。主要是看laser_avoidance.py文件。

laser_avoidance.py

主要看雷达的回调函数，这里说明了如何获取到每个角度的障碍物距离信息，

```

def registerScan(self, scan_data):
    if not isinstance(scan_data, LaserScan): return
    self.right_warning = 0
    self.left_warning = 0
    self.front_warning = 0
    ranges = np.array(scan_data.ranges)
    for i in range(len(ranges)):
        if ranges[i] < self.ResponseDist: #range[i]就是雷达扫描的结果，这里指的是
距离信息
            angle = (scan_data.angle_min + scan_data.angle_increment * i) *
180 / pi
            if angle > 180: angle = angle - 360 #angle是根据雷达的结构来设定判断
范围的
            if -self.LaserAngle < angle < -20:
                self.right_warning += 1
            elif abs(angle) <= 20:
                self.front_warning += 1
            elif 20 < angle < self.LaserAngle:
                self.left_warning += 1

    if self.Joy_active == True or self.Switch == False:
        if self.Moving == True:
            self.pub_vel.publish(Twist())
            self.Moving = not self.Moving
        return

    self.Moving = True

```